

This three branches produce three different types of features of same dimensional tensor and Here CGAF adaptively weight their contribution. Standard fusion techniques like **Concatenation** or **Cross-Attention** operate on a flawed assumption: that every deepfake leaves the same kind of 'fingerprint' . But in reality, forgery traces are domain-specific.

For eg

- 1) **NeuralTextures** artifacts are strongest in the **Spatial** domain;
- 2) **Deepfakes** and **FaceShifter** are revealed by **Frequency** artifacts; and
- 3) **FaceSwap** leaves **Noise** inconsistencies at blending boundaries.

A static weighted sum is suboptimal because the importance of these domains varies per sample. We needed a system that doesn't just 'add' data, but 'interrogates' it." For this we make our own mechanism that is CGAF(

1: The Consistency Check"Our first stage is the **Consistency Check**. Before we fuse anything, we project our features and calculate **Cosine Similarity Maps**. The logic is simple: we look for agreement. If both the Spatial and Frequency branches detect an anomaly in the same region, the model gains mathematical confidence that a forgery is actually present."

2: Gated Filter & Fallback "Second, we use **Temperature-Scaled Sigmoid Gating**. Unlike standard attention, we use **Log-Parameterization** to keep our temperature bounded between **0.1 and 10.0**. This prevents 'dead gradients' and ensures the model keeps learning.

Crucially, we've implemented a **Conservative Fallback**. We use **Cross-Consistency Modulation**: if the Frequency and Noise branches disagree—represented by a low C_{fn} —the model suppresses those gates. It falls back to the Spatial branch as a 'safe' baseline, preventing 'garbage' signals from ruining the final prediction."

3: Global Semantic Oversight "Finally, we have the 'brain' of the module: **Global Semantic Oversight**. We use high-level features—**F_high**—to generate a global gating signal.

- If the scene looks stable and natural, this signal suppresses the noise and frequency gates to **reduce false positives**.
- If the global context indicates manipulation, it **amplifies** those auxiliary branches to catch even the most subtle artifacts.

Overall CGAF give better result than traditional modules because it doesn't just "add" these domains; it **interrogates** them using a multi-step gated logic

"Once our **CGAF** module has fused the spatial, frequency, and noise data for each frame, we are left with a rich feature set. But deepfakes aren't just static images—they are videos. To catch them, we must look at how these features change over time."

For this we use temporal modelling. First stage of this module is

1. Masked Spatial Pooling "Instead of using basic averaging, which can be 'polluted' by background pixels, we use **Energy-Weighted Masked Pooling**. here calculate the feature energy for every pixel. This acts like a soft attention mask, forcing the model to focus on high-activity areas—like the face—while suppressing irrelevant background noise."

2. Multi-Scale Temporal Differences]"This is where we address a major weakness of single-frame detectors. Video deepfakes often exhibit **temporal flickering** and motion inconsistencies that are invisible in a still photo. We compute frame differences at three distinct temporal scales:

- **Scale 1 ($\Delta 1$):** Captures high-speed jitter and fast GAN inconsistencies.
- **Scale 2 ($\Delta 2$):** Detects short-range temporal drift.
- **Scale 4 ($\Delta 4$):** Identifies medium-range motion errors. By analyzing these three scales, we ensure no artifact—whether it's a quick flash or a slow drift—goes unnoticed."

3. Sequence Construction & Transformer Encoding "All of these features—the original frame data and our multi-scale differences—are concatenated into a master sequence.

- **Motion-Aware Bias:** We add a specific 'motion-aware bias' to every token to tell the model which frames contain the most movement.
- **The Transformer:** This sequence is fed into a **2-layer Transformer Encoder**. This is the final decision-maker. It uses multi-head attention to compare every frame and every difference across the entire video, producing a single, high-dimensional 'video-level feature' that summarizes the forgery evidence."

"By combining **CGAF's** multi-domain analysis with this **Multi-Scale Temporal Difference** encoding, we've built a model that is sensitive to both the microscopic artifacts of AI generation and the macroscopic inconsistencies of video motion."

The final stage of the **AFAGNet v3** architecture moves from feature extraction to the actual decision-making and visual explanation. This is handled by two specialized heads that provide both a final verdict and a precise map of the forgery.

Classification and Localization: The Final Verdict

1. Classification Head: Context-Aware Decision Making The classification head is responsible for the final 'Real or Fake' decision. It doesn't just look at the temporal features ; it also integrates high-level semantic context.

- **Content-Aware Gating:** We use a learnable gate to determine exactly how much the semantic context should influence the final result.
- **Progressive Dropout:** To ensure the model generalizes well and doesn't overfit to specific datasets, we use a 4-layer MLP with progressive dropout—starting at **0.3** and reaching **0.4** in the middle layers.
- **#####Numerical Stability:** The head returns raw logits during training for better numerical stability, while the final sigmoid is applied only during inference." #####

2. Localization Head: Mapping the Forgery]"One of the strongest features of our model is that it doesn't just say 'this is a fake'—it shows you where. The localization head decodes the fused features back into a 224x224 forgery probability mask.

- **Semantic Skip Connections:** To keep the spatial details sharp, we include a skip connection from F_{high} . This ensures that as we upsample the features, we don't lose the high-level semantic context of the face.
- **Progressive Upsampling:** We use a cascade of **Transposed Convolutions** to step up the resolution from 14 x14 all the way to 224 xx224
- **Middle-Frame Focus:** While we generate masks for the entire sequence, we prioritize the middle frame ($N//2$) as our primary output to ensure we are capturing the peak of the temporal information."

3. Explainability: Transparency in AI "Finally, we've built in **Explainability** features that go beyond standard detection:

- **Confidence Maps:** We calculate exactly how confident the model is about each pixel, where values closer to **0** or **1** represent higher certainty.
- **Temporal Stability:** We measure how much the prediction changes from frame to frame. A stable mask across time suggests a more reliable detection of a consistent forgery.
- **Domain Attribution:** We can even trace the detection back to its source—telling us whether the spatial, frequency, or noise branch was the primary driver for the decision."

"By combining high-level classification with pixel-perfect localization and explainability, our model provides a transparent and robust defense against modern deepfake technology."

The training strategy for **AFAGNet v3** is designed to handle real-world challenges like class imbalance and missing data while ensuring the model remains stable during optimization.

1. The Multi-Task Loss Function

The total loss is a weighted combination of four specialized objectives:

- **Classification Loss** : We use **Focal Loss** instead of standard Binary Cross-Entropy (BCE). This is critical because our dataset has a **1:4.6 imbalance** (mostly fakes). Focal loss prevents the model from simply guessing "fake" every time and forces it to focus on "hard" samples—high-quality deepfakes that are difficult to distinguish.
- **Localization Loss** : To learn where the forgery is, we use a mix of ground-truth masks and **pseudo-masks**. Pseudo-masks are generated from the Frequency and Noise branches, allowing the model to learn localization even for datasets like FaceShifter that don't provide ground-truth masks.
- **Consistency Losses**:
 - **Branch Consistency** : Encourages the Spatial, Frequency, and Noise branches to semantically align.
 - **Temporal Consistency** : Uses a **clamped** difference between consecutive frame masks to prevent unstable frames from causing "exploding gradients" during training.

2. Scheduled Supervision

We use a **scheduled decay** for the localization weight λ_{loc} .

- **Early Training**: We use high spatial supervision (0.8) to help the model find forgery regions.
- **Late Training**: We decay this weight down to 0.1 . This prevents the localization task from "overpowering" the classification task, ensuring the final "Real vs. Fake" verdict remains accurate.

3. Optimization & Stability

To ensure the model converges effectively on hardware like the RTX 3050, we use **AdamW** with a specialized learning rate strategy:

- **Differential Learning Rates:** The pretrained MobileViT-v2 backbone is trained at **10%** of the base learning rate to preserve its existing features, while the new modules (like CGAF) train at the full rate.
- **Warmup & Cosine Annealing:** We start with a 2-epoch linear warmup to stabilize the initial gradients, followed by a cosine decay to fine-tune the weights.
- **Exponential Moving Average (EMA):** We maintain a "smoothed" version of the model weights (0.9999 decay). This typically provides a **0.5–1.5% AUC improvement** by averaging out noise in the training trajectory.

"Our training isn't just about minimizing error; it's about balance. By using Focal Loss for class imbalance, pseudo-masks for missing data, and EMA for stability, we ensure AFAGNet v3 is as robust in training as it is in inference."